

Software Evolution — Phase 2 Report

Link <https://github.com/ahmedramy9/unitime/pull/5>

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1. What We Did

UniTime is an open-source university timetabling system. For this phase we picked a usability gap in the Class Search page: when a user ran a search, the page showed the results table but gave no indication of how many classes matched. We added a count message that appears just above the table after every successful search — for example, *Found 12 classes* — and disappears when there are no results.

Three files were changed, adding eight lines in total. No existing logic was removed and no database changes were made.

2. Files Changed

File	Change
<code>ClassSearchAction.java</code>	Added <code>request.setAttribute("resultCount", classes.size())</code> in both the automatic and manual search paths (lines 200 and 306)
<code>CourseMessages.java</code>	Added <code>@DefaultMessage("Found {0} classes") String foundClasses()</code>
<code>classSearch.jsp</code>	Added a conditional block that reads the count from the request and renders it above the table

```
<s:if test="showTable == true && #request.resultCount != null">
  <div style="padding:8px; background:#f0f0f0;">
    <strong>
      <loc:message name="foundClasses">
        <s:param value="%{#request.resultCount}"/>
      </loc:message>
    </strong>
  </div>
</s:if>
```

3. UML Diagrams

Three diagrams were reverse-engineered from the UniTime source code, Hibernate mappings, and configuration files.

3.1 System Component Diagram

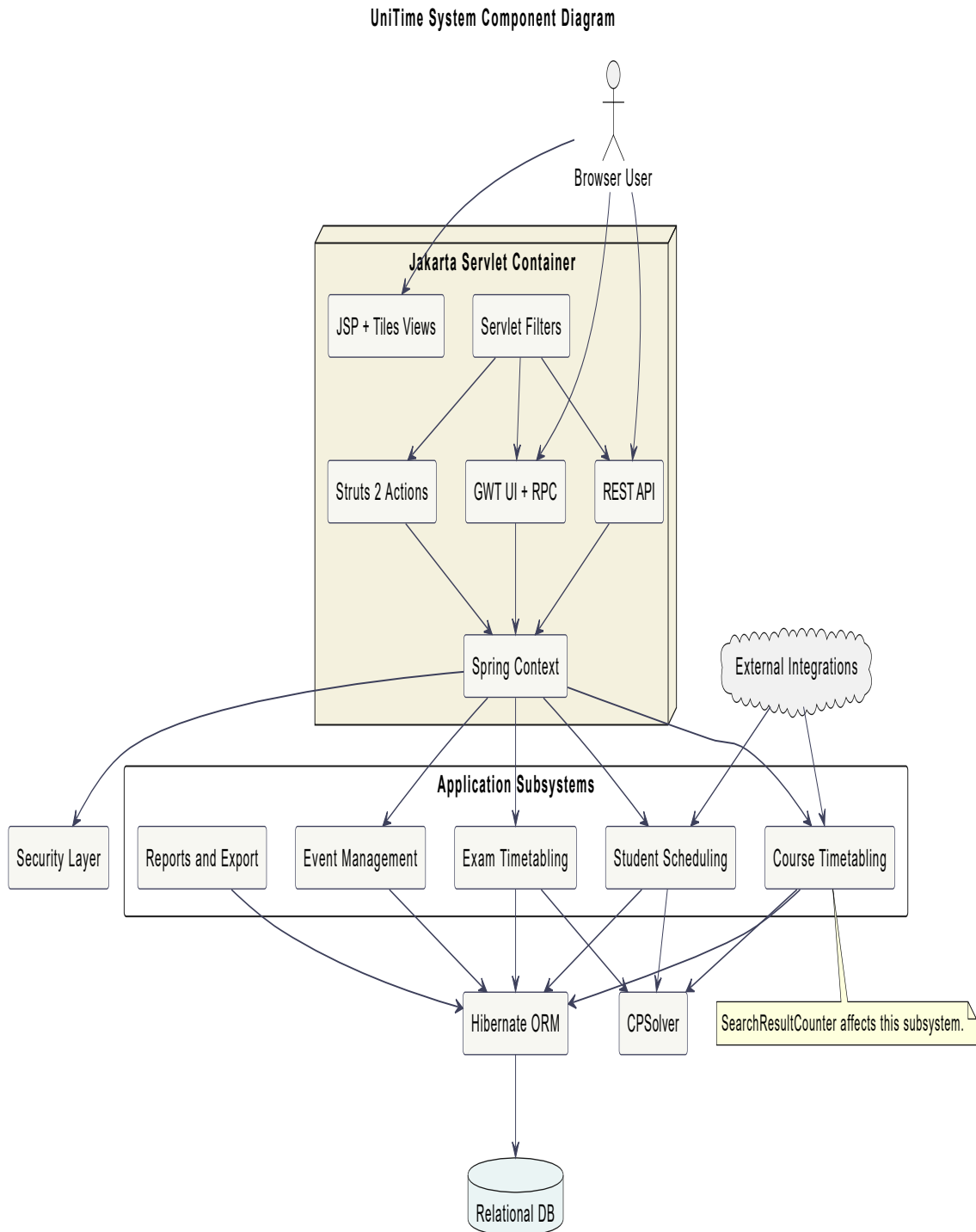


Figure 1 — UniTime system component diagram

3.2 Class Search Subsystem — Class Diagram

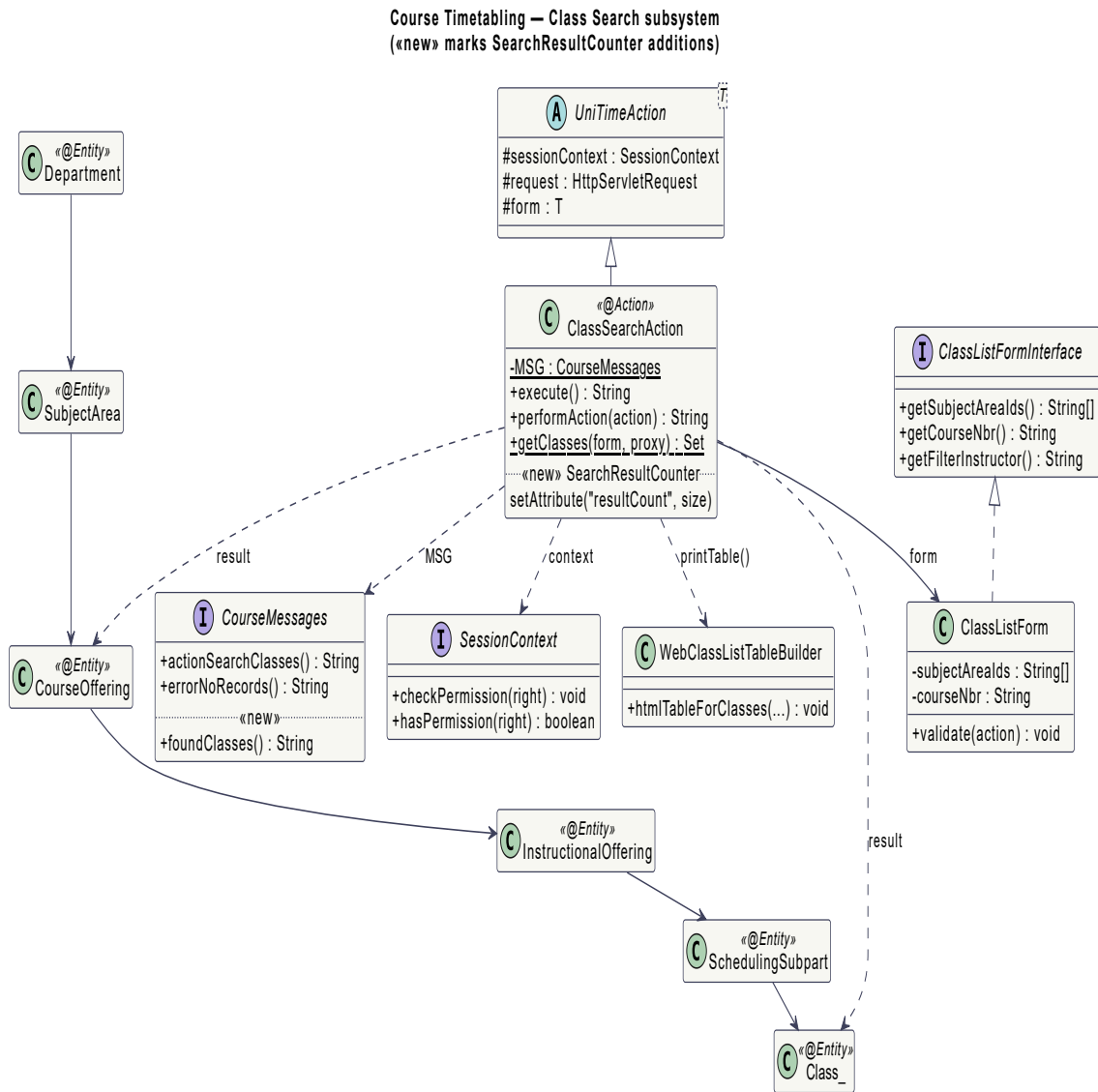


Figure 2 — Course Timetabling / Class Search class diagram. Elements marked [new] were added by this change.

3.3 Class Search Request Flow — Sequence Diagram

Class Search Request Sequence

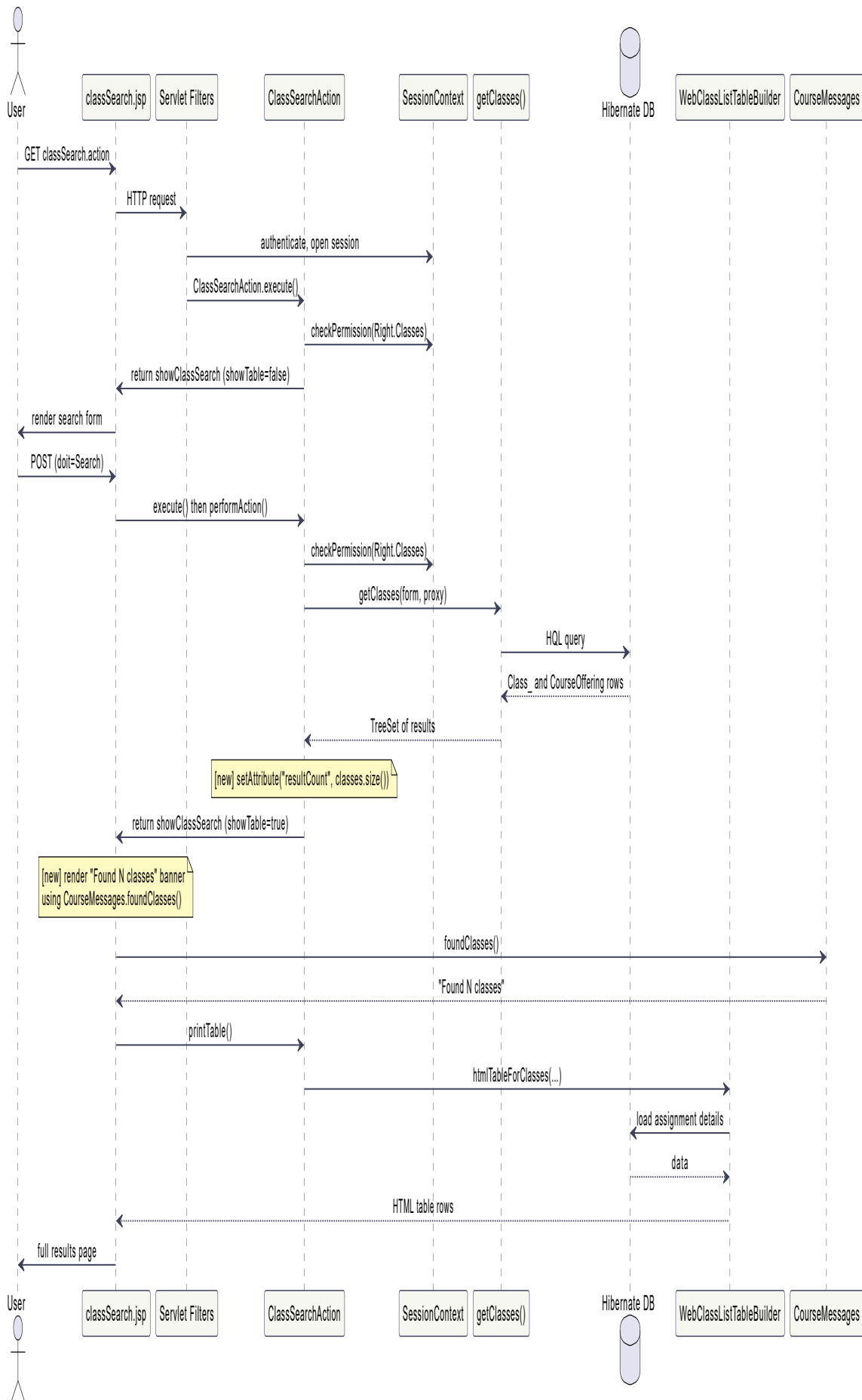


Figure 3 — Class Search sequence diagram. Steps marked [new] were added by this change.

3.4 Before and After

Element	Before	After
<code>ClassSearchAction</code>	Fetches classes, returns view	Also stores result count in request scope
<code>CourseMessages</code>	No <code>foundClasses()</code>	New method returning "Found {0} classes"
<code>classSearch.jsp</code>	Table only	Count banner above table

4. Impact Analysis

4.1 Starting Impact Set (SIS)

The two most obvious components to check first based on the change request:

```
SIS = { ClassSearchAction.java, classSearch.jsp } |SIS| = 2
```

4.2 Candidate Impact Set (CIS)

Tracing dependencies from the SIS added two more candidates:

- `CourseMessages.java` — already used by the JSP for user-facing text, a new message key is needed.
- `ClassListForm.java` — holds the classes collection, included as a precaution.

```
CIS = { ClassSearchAction.java, classSearch.jsp, CourseMessages.java,
ClassListForm.java } |CIS| = 4
```

4.3 Actual Impact Set (AIS)

`ClassListForm.java` did not need modification. The count was available directly from the existing collection.

```
AIS = { ClassSearchAction.java, classSearch.jsp, CourseMessages.java }
|AIS| = 3
```

4.4 DIS and FPIS

```
DIS = { } |DIS| = 0 FPIS = { ClassListForm.java } |FPIS| = 1
```

4.5 Metrics

Metric	Formula	Value	Notes
Recall	$ CIS \cap AIS / AIS $	1.0	All modified files were predicted. DIS is empty.
Precision	$ CIS \cap AIS / CIS $	0.75	One false positive (ClassListForm).
Inclusiveness	$AIS \subseteq CIS$	1	No actual change was missed.
Amplification	$ IISO / DISO $	0	No ripple effect.
ChangeRate	$ CIS / System $	0.0016	4 out of ~2500 system objects.
S-Ratio	$ AIS / CIS $	0.75	Good fit between prediction and reality.

4.6 Ripple Effect

No ripple effect was introduced. The `resultCount` attribute is write-only and does not feed back into any search logic. The `foundClasses()` method is new so no existing code references it. No new classes, queries, or relationships were added.

4.7 Change Propagation

Step	Component	Action
1	<code>ClassSearchAction.java</code>	Stores result count in HTTP request scope
2	<code>CourseMessages.java</code>	Defines the localized display message
3	<code>classSearch.jsp</code>	Renders the count banner above the table
4	—	No further propagation

4.8 Regression Testing Scope

Component	What to test
<code>ClassSearchAction.java</code>	Both search paths set <code>resultCount</code> correctly
<code>classSearch.jsp</code>	Banner shows when results exist, hidden when empty, exports unaffected
<code>CourseMessages.java</code>	Message formats correctly with a numeric parameter

5. Verification

Check	Result
<code>setAttribute("resultCount", ...)</code> in automatic search path (line 200)	✓
<code>setAttribute("resultCount", ...)</code> in manual search path (line 306)	✓
<code>foundClasses()</code> added to <code>CourseMessages.java</code> (line 1948)	✓
Conditional block in <code>classSearch.jsp</code> (lines 341–345)	✓
Search with results → count banner appears	✓
Search with no results → no banner	✓
PDF / CSV export → unaffected	✓

6. Git

Commit	Description
<code>92c1bfa</code>	Add SearchResultCounter feature
<code>f533027</code>	Add UML diagrams and impact analysis
<code>17781b0</code>	Rewrite impact analysis to match formal methodology
<code>5b671bc</code>	Clean up Documentation/UML folder

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